

Applying FAIR4RS Principles to Research Software Using Codefair

Bhavesb Patel, Ph.D.
Research Professor (presenter)

Dorian Portillo, B.S.
Software Developer

Sanjay Soundarajan, M.S.
Software Developer





Making Software FAIR

Making Software FAIR

What is FAIR?

FAIR (Findable, Accessible, Interoperable, Reusable) Principles (2016):
High-level recommendations to make research outcomes reusable

To be Findable:

- F1. (meta)data are assigned a globally unique and persistent identifier
- F2. data are described with rich metadata (defined by R1 below)
- F3. metadata clearly and explicitly include the identifier of the data it describes
- F4. (meta)data are registered or indexed in a searchable resource

(and more)

<https://doi.org/10.1038/sdata.2016.18>

Making Software FAIR

FAIR4RS principles

FAIR Principles for Research Software or FAIR4RS Principles (2022)
provide instructions for making software findable and reusable

F1. Software is assigned a globally unique and persistent identifier.
F1.1. Components of the software representing levels of granularity are assigned distinct identifiers.
F1.2. Different versions of the software are assigned distinct identifiers.
F2. Software is described with rich metadata.
F3. Metadata clearly and explicitly include the identifier of the software they describe.
F4. Metadata are FAIR, searchable and indexable.

(and more)

<https://doi.org/10.1038/s41597-022-01710-x>

Making Software FAIR

FAIR4RS principles – why?



Having a unique identifier → Making your software citable



Having rich metadata → Making your software findable

Making Software FAIR

Challenge

The FAIR4RS Principles are **not actionable** by design

F1. Software is assigned a globally unique and persistent identifier.

How do I assign a unique identifier?

F1.1. Components of the software representing levels of granularity are assigned distinct identifiers.

F1.2. Different versions of the software are assigned distinct identifiers.

F2. Software is described with rich metadata.

What is rich metadata?

F3. Metadata clearly and explicitly include the identifier of the software they describe.

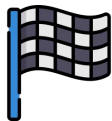
F4. Metadata are FAIR, searchable and indexable.

(and more)

<https://doi.org/10.1038/s41597-022-01710-x>

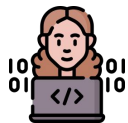
Making Software FAIR

FAIR-BioRS (2023): Actionable guidelines



Before starting

- Work from a version controlled system platform (e.g. GitHub)
- Select a license and include a LICENSE file



While developing the software

- Record dependencies (e.g. requirements.txt for Python)
- Maintain a README



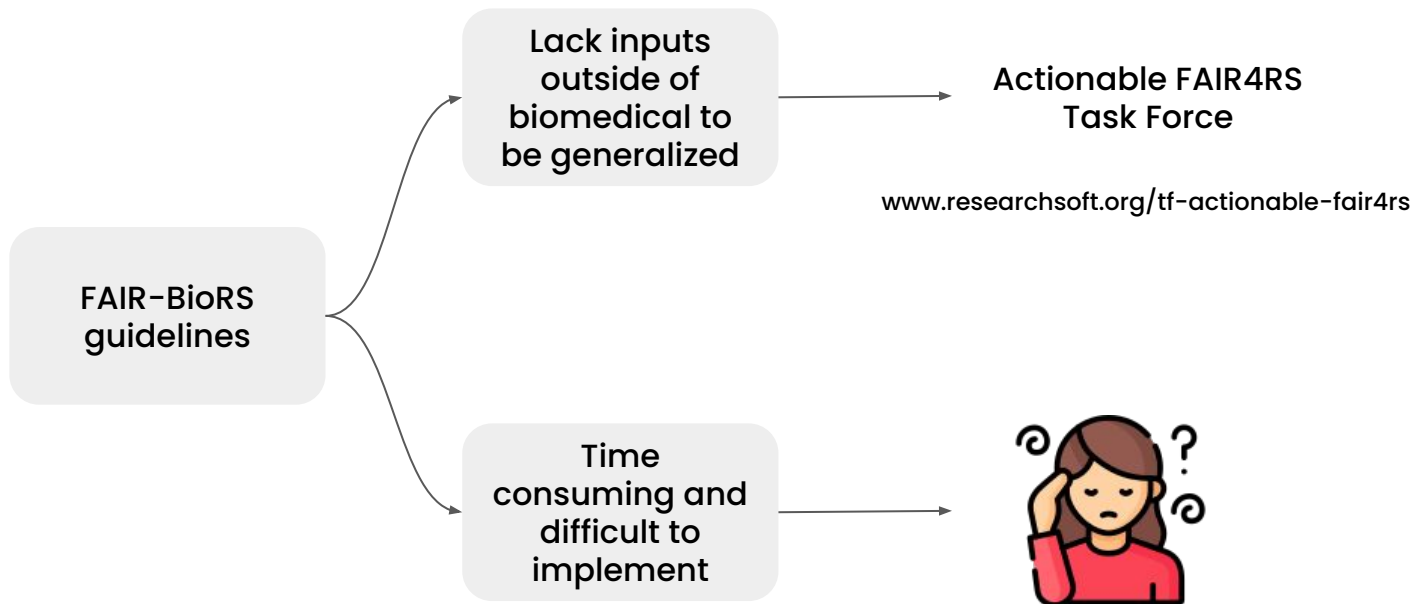
When releasing a new version

- Include metadata in codemeta.json, CITATION.cff, CHANGELOG
- Archive source code on a DOI-issuing repository like Zenodo

Full guidelines: <https://doi.org/10.1038/s41597-023-02463-x>

Making Software FAIR

More challenges





Codefair

Codefair

Overview

Open source and free platform that acts as your
personal assistant for making software FAIR at the source



Started in 2024 as a side project mostly to help our team



Received \$20k in Azure credits from the Microsoft AI for Good Lab



We chose GitHub as our first integration point

Codefair

Demo

The screenshot shows a GitHub repository interface. At the top, the breadcrumb navigation reads 'FAIR-BioRS / my-data-analysis-code'. A search bar is present with the placeholder 'Type / to search'. Below this is a navigation bar with links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. The repository name 'my-data-analysis-code' is displayed with a 'Public' badge. Interaction buttons for 'Edit Pins', 'Watch' (0), 'Fork' (0), and 'Star' (0) are visible. The main content area shows the 'main' branch with '1 Branch' and '0 Tags'. A commit history table lists recent changes: 'bvhpatel' updated README.md (4 commits, now), 'LICENSE' was the initial commit (7 minutes ago), 'README.md' was updated (now), and 'code.py' was added via upload (1 minute ago). Below the commit history, the 'README' tab is active, showing the title 'My data analysis code' and a link to 'Repo to demo codefair.io'. On the right, the 'About' section provides repository statistics: 0 stars, 0 watching, and 0 forks, along with links to the README, MIT license, and activity. The 'Releases' and 'Packages' sections at the bottom indicate that no releases or packages have been published yet.

FAIR-BioRS / my-data-analysis-code

Q Type / to search

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

my-data-analysis-code Public

Edit Pins Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file t + <> Code

bvhpatel Update README.md f2c121c · now 4 Commits

LICENSE	Initial commit	7 minutes ago
README.md	Update README.md	now
code.py	Add files via upload	1 minute ago

README MIT license

My data analysis code

Repo to demo [codefair.io](#)

About

Repo to demo codefair.io

- Readme
- MIT license
- Activity
- Custom properties
- 0 stars
- 0 watching
- 0 forks

Report repository

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

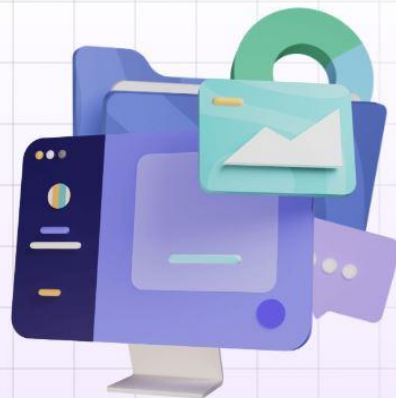
Codefair

Demo

[About](#)[FAIR Software](#)[Dashboard](#)

Make your research software reusable without breaking a sweat!

Let Codefair handle the complexities of making your research software FAIR, so you can focus on what matters most to you.

[Get Codefair →](#)[Read the Documentation](#)

Codefair

Demo

 Marketplace / Apps

 **codefair.io** App Add ...

README Transparency

Codefair is your personal assistant for making research software reusable and compliant with the [Findable, Accessible, Interoperable, Reusable \(FAIR\) Principles for Research Software](#). Whether you're developing AI/ML models with Python, data visualization tools with Jupyter, or data analysis code with R, Codefair is here to help.

Make your research software reusable without breaking a sweat!

By combining a GitHub issue dashboard, a website with intuitive user interfaces, and automated pull request, Codefair ensures your software follows best practices, provides standard metadata, includes a license file, is archived on Zenodo, and more. All you need to do is:

1. Install codefair-app on the GitHub organizations/repositories of your choice (click "Install it for free" below)
2. Code as usual
3. Track FAIR compliance issues through the issue dashboard and address them through the Codefair website

With Codefair by your side, you're not just developing software but you're advocating for better software practices.

Note

While Codefair is free, installing it via the GitHub Marketplace may still require a credit card to be associated with your (or your organization's) GitHub account. To circumvent this, you can alternatively install Codefair directly from the [app page](#).



About

Codefair is your personal assistant when it comes to making your research software reusable

By [fairdatahub](#)  40 installs

Verified

GitHub has verified the publisher's identity, ownership of their domain, and compliance with [other requirements](#).

Tags

[code-quality](#) [utilities](#)

Pricing

Basic plan available.

Resources

-  Support
-  Documentation
-  Privacy Policy
-  Terms of Service
-  Report abuse

Codefair

Demo

- Archive
- Logs
- Deleted repositories
- < > Developer settings

✓ Read access to metadata

✓ Read and write access to code, issues, and pull requests

Repository access

☐ All repositories

This applies to all current and future repositories owned by the resource owner. Also includes public repositories (read-only).

☒ Only select repositories

Select at least one repository. Also includes public repositories (read-only).

Select repositories -

Selected 2 repositories.

FAIR-BioRS/my-data-analysis-code

FAIR-BioRS/Hub

Save

Cancel

Danger zone

Suspend your installation

This will block the app access to your resources.

Suspend

Uninstall "codefair.io"

Codefair Demo

The screenshot shows the GitHub interface for the repository 'FAIR-BioRS / my-data-analysis-code'. The 'Issues' tab is active, displaying the 'FAIR Compliance Dashboard #1'. The dashboard is a checklist for making software FAIR. It includes sections for README, LICENSE, Metadata, and FAIR Software Release. The Metadata section is highlighted with a red box. The right sidebar shows settings for the issue, including assignees, labels, type, projects, milestones, relationships, development, notifications, and participants.

FAIR Compliance Dashboard #1

codefair-io bot opened now

Check the FAIRness of your software

This issue is your repository's dashboard for all things FAIR. Keep it open as making and keeping software FAIR is a continuous process that evolves along with the software. You can read the [documentation](#) to learn more.

README ✓

A `README.md` file was found at within your repository.

[Edit README](#)

LICENSE ✓

A `LICENSE` file is found at the root level of the repository.

[Edit License](#)

Metadata ✗

To make your software FAIR, a `CITATION.cff` and `codemeta.json` are expected at the root level of your repository. These files are not found in the repository. If you would like Codefair to add these files, click the "Add metadata" button below to go to our interface for providing metadata and generating these files.

[Add Metadata](#)

FAIR Software Release ✗

To make your software FAIR, it is necessary to archive it in an archival repository like Zenodo every time you make a release. When you are ready to make your next release, click the "Create release" button below to easily create a FAIR release where your metadata files are updated (including with a DOI) before creating a GitHub release and archiving it.

[Create Release](#)

Assignees: No one - [Assign yourself](#)

Labels: No labels

Type: No type

Projects: No projects

Milestone: No milestone

Relationships: None yet

Development: [Code with agent mode](#)

[Create a branch](#) for this issue or link a pull request.

Notifications: [Subscribe](#)

You're not receiving notifications from this thread.

Participants: No participants

→ Transfer issue

🔄 Duplicate issue

Codefair

Demo

Edit metadata for [my-data-analysis-code](#) [Need help?](#)

To make your software FAIR, a CITATION.cff and codemeta.json file are expected at the root level of your repository. They help people discover your software and provide information about your software to them. Provide metadata about your software below and codefair will submit a pull request with a CITATION.cff and codemeta.json file for you.

> Basic Information
General information of the repository.

Software Name *

my-data-analysis-code

Description *

Repo to demo codefair.io

Creation Date

2025-09-30

First Release Date

Select Date

> Authors and Contributors
Information about the authors and contributors of the software.

Authors *

Bhavesht Patel

Remove Author

^



Codefair

Demo

Default Value

Download URL

Current Version Release Notes

> Additional Information

Additional information about the software.

Development Status

Is Source Code Of

Is Part Of

 Save draft

 Save and push to repository



Codefair

Demo

FAIR-BioRS / my-data-analysis-code

feat: ✨ Add code metadata files #2

codefair-io wants to merge 2 commits into main from code-metadata-0t@qs

Files changed: 2

Filter changed files

CITATION.cff

codemeta.json

```
@@ -0,0 +1,14 @@
1 + title: my-data-analysis-code
2 + authors:
3 +   - given-names: Bhavesh
4 +     family-names: Patel
5 +     affiliation: California Medical Innovations Institute
6 + cff-version: 1.2.0
7 + message: If you use this software, please cite it using the metadata from this file.
8 + type: software
9 + abstract: Repo to demo codefair.io
10 + keywords:
11 +   - python
12 +   - biomedical
13 + license: MIT
14 + repository-code: https://github.com/FAIR-BioRS/my-data-analysis-code
```

```
@@ -0,0 +1,30 @@
1 + {
2 +   "name": "my-data-analysis-code",
3 +   "@context": "https://w3id.org/codemeta/3.0",
4 +   "author": [
5 +     {
6 +       "affiliation": {
7 +         "name": "California Medical Innovations Institute",
8 +         "type": "Organization"
9 +       },
10 +       "familyName": "Patel",
11 +       "id": "_author_1",
12 +       "givenName": "Bhavesh",
13 +       "type": "Person"
14 +     }
15 +   ]
16 + }
```

Codefair

Demo

  FAIR-BioRS / my-data-analysis-code

 Code  Issues 1  Pull requests 1  Actions  Projects  Wiki  Security  Insights  Settings

Type to search      

FAIR Compliance Dashboard #1  



 codefair-io  opened 5 minutes ago · edited by codefair-io  Contributor 

Check the FAIRness of your software

This issue is your repository's dashboard for all things FAIR. Keep it open as making and keeping software FAIR is a continuous process that evolves along with the software. You can read the [documentation](#) to learn more.

README ✓

A `README.md` file was found at within your repository.

[Edit README](#)

LICENSE ✓

A `LICENSE` file is found at the root level of the repository.

[Edit License](#)

Metadata ✓

A `CITATION.cff` and `codemeta.json` file are found in the repository. They may need to be updated over time as new people are contributing to the software, etc.

[Edit Metadata](#)

FAIR Software Release ✗

To make your software FAIR, it is necessary to archive it in an archival repository like Zenodo every time you make a release. When you are ready to make your next release, click the "Create release" button below to easily create a FAIR release where your metadata files are updated (including with a DOI) before creating a GitHub release and archiving it.

[Create Release](#)

Additional Recommendations

Assignees 

No one - [Assign yourself](#)

Labels 

No labels

Type 

No type

Projects 

No projects

Milestone 

No milestone

Relationships 

None yet

Development 

 Code with agent mode 

[Create a branch](#) for this issue or link a pull request.

Notifications [Customize](#)

 [Subscribe](#)

You're not receiving notifications from this thread.

Participants

No participants

[Transfer issue](#)

[Duplicate issue](#)

Codefair

Future work



Promote Codefair (currently installed on 760+ repo across 40 users/orgs)



Add more features (README improvement, code formatting, CHANGELOG) and **more integrations** (GitLab, CI/CD)



Align with the Actionable FAIR4RS guidelines when they are available



Get funding to move this from a side project to an actively developed and sustainable platform

Thank You!



bpatel@calmi2.org



fairdataihub.org

*Find these slides and
all resources here*



bit.ly/codefair-hack

This work is supported by Azure sponsorship credits granted by Microsoft's AI for Good Research Lab



Hands-on Demo

Hands-on Demo

- 1 Go to codefair.io
- 2 Install the Codefair GitHub app on one of your repos
- 3 Add CITATION.cff and CodeMeta files to your repo